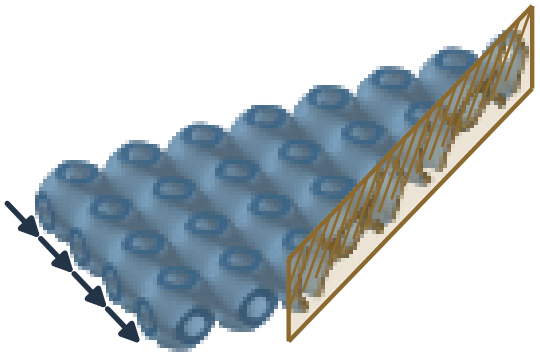
Cut TPMS lattice, clamped on its cut



per cell: cell-face and cut-plane DOFs retained, interior condensed

NICE: one condensed operator per cell

learned extension → **two-grid** correction → **energy** form $\widehat{\Psi}^T K \widehat{\Psi}$

learning: trial field | **correction: improvability** | **variational form: structure**

Clamped cut layer

homogenisation under-estimates compliance by

27%

NICE matches exact to

0.03%

(gradient 0.3%)